

HARSH SHROFF

Open to Relocation

☎ 667-464-5255 ✉ harshshroff@gmail.com

🌐 [linkedin.com/in/harshshroff/](https://www.linkedin.com/in/harshshroff/) 🐙 github.com/HarshShroff/ 🔗 harshshroff.github.io/

SUMMARY

AI/ML Engineer and Data Scientist with a strong foundation in generative AI, deep learning, and end-to-end ML pipeline development. Proven success delivering production-grade solutions in domains including geospatial analytics, robotics, and aquaculture, with hands-on expertise in LLMs, AWS, MLOps frameworks, and real-time computer vision systems. **AWS Machine Learning Associate Certified**, with a track record of automating complex workflows, improving decision-making efficiency, and publishing research in peer-reviewed journals. Adept at building scalable AI systems from prototype to deployment using tools like PyTorch, LangChain, YOLOv8, Streamlit, and Vertex AI. I bring a rare blend of technical depth, cross-domain adaptability, and a builder mindset to drive real-world impact through intelligent systems.

SKILLS

Machine Learning & AI: Generative AI (*LLMs, RAG, LangChain, LlamaIndex*), Deep Learning (*TensorFlow, PyTorch, YOLOv8, OpenCV*), Natural Language Processing (*NLP*), Computer Vision (*CV*), Time Series Forecasting, Predictive Modeling, MLOps (*MLflow, Kubeflow, Vertex AI*), Prompt Engineering, RAG (*Retrieval-Augmented Generation*)

Data Engineering & Cloud Computing: AWS (*Lambda, EC2, S3, Textract, Comprehend, Glue, Redshift, Kinesis, Bedrock*), SQL & NoSQL Databases (*PostgreSQL, MySQL, MongoDB, Elasticsearch*), Data Pipelines (*ETL, Airflow, Terraform, Spark, Hadoop*)

Data Analysis & Visualization: Python (*Pandas, NumPy, SciPy, Seaborn, Matplotlib*), BI Tools (*Tableau, Power BI, Streamlit, Dash*), Statistical Analysis (*A/B Testing, Regression, Hypothesis Testing*), CI/CD, Git Version Control

Software Development & Automation: Python, R, SQL, C++, Embedded C, API Development (*FastAPI, Flask*), DevOps (*CI/CD, Docker, Kubernetes, Terraform*), Agile Methodologies, Git, Version Control

EXPERIENCE

ML Engineer

Jun 2024 – Present

VITG Corp., Halethorpe, MD

- Developed a resume scoring Slack bot powered by Llama LLM and AWS services (Lambda, EC2, S3) to automate candidate screening, reducing manual processing by 60%.
- Built a chatbot-driven geospatial tool to match addresses with government conservation zones, aiding environmental planning and property regulations in Hyderabad.
- Automated geospatial data extraction from 2500+ scanned PDFs/images using OCR-based lat-long extraction and Claude 3.5 Haiku LLM, visualizing results on interactive maps.
- Designed and deployed an AI-powered web-app using Streamlit and AWS for real-time geospatial analysis, improving data accessibility by 30%.

AI Researcher

Mar 2023 – Dec 2024

Center for Real-time Distributed Sensing and Autonomy, UMBC

- Led a team of 5 to develop AI-powered multi-modal robotic navigation for Boston Dynamics Spot, Clearpath Husky & Jackal robots under the ArtIAMAS project (funded by the US Army Research Lab & DEVCOM).
- Implemented gesture-based robotic controls using Microsoft HoloLens 2 and Python, enabling intuitive human-robot interaction for the "Human Machine Teaming" project.
- Optimized LIDAR-based pathfinding algorithms using reinforcement learning, reducing navigation errors by 20% in dynamic environments.
- Designed a GPS-based navigation system for Clearpath Husky robots, achieving 95% path accuracy.

Data Science Intern

Aug 2023 – Dec 2023

Freshwater Institute – The Conservation Fund, Shepherdstown, WV

- Achieved a 92% accuracy rate in fish fillet color grading using Python, OpenCV & YOLOv8 Classifier model deployed on Raspberry Pi 4.
- Designed and developed an in-house 3D printed portable case for a real-time assessment on fillet grading in aquaculture.
- Co-authored a research paper titled "Fish Fillet Color Profiling Using Deep Learning", detailing the application of AI in precision aquaculture and food safety.

EDUCATION

Master's in Data Science

May 2024

University of Maryland Baltimore County (UMBC)

- GPA: 3.8/4.0
- Relevant Coursework: Natural Language Processing, Data Analysis & Visualization, Data Management, Big Data Processing, Financial Data Science, Machine Learning, Artificial Intelligence

Bachelor's in ELectronics & Communication Engineering

May 2022

Gujarat Technological University (GTU)

- GPA: 3.8/4.0 – *converted*
- Relevant Coursework: Introduction to Python, Introduction Machine Learning & Artificial Intelligence

PROJECTS

Document Parsing for Environmental Planning

AWS (Textract, S3, Redshift), Python, Tableau

- Automated geospatial data extraction from 2500+ scanned PDFs using OCR and ML models, increasing data processing efficiency by 40%.
- Developed prompt engineering strategies to refine AI outputs for conservation project prioritization.
- Designed interactive Streamlit dashboard, providing stakeholders with real-time insights, and enabling faster decision-making on conservation projects.

Dashboard for Autonomous Robotic Patrols

Python, ROS, LiDAR, Deep Learning, Computer Vision

- Developed a real-time asset monitoring dashboard using Python, Dash and ROS, tracking battery health, network connectivity, and asset movement via GPS in military simulations.
- Integrated real-time object detection using OpenCV and Python and obstacle detection via LiDAR, enhancing robotic autonomous movement and safety.

AirbnbLense – Personalized Recommendation System ([GitHub](#))

Python, OpenAI API, Computer Vision, Tableau, Big Data

- Built a recommendation engine combining structured and unstructured data (160MBs of CSV, 130GBs of images).
- Implemented computer vision models to analyze listing images for style, color, and object recognition.
- Reduced Airbnb bounce rates by 20% and improved engagement by 40% through tailored recommendations.

Cyclists' Performance Dashboard ([GitHub](#))

Python (Dash, Seaborn), Machine Learning

- Developed an interactive dashboard to track real-time cyclist performance metrics using Python and Dash, improving training efficiency by 20%.
- Implemented predictive analytics for personalized training insights, enhancing athlete performance.

CERTIFICATIONS & PUBLICATIONS

Certifications

- **AWS Machine Learning Associate Certification** – [View Badge](#)
- **AWS Machine Learning Foundations Certification** – Proficiency in core ML concepts & AWS tools.

Publications

- Published research in *IEEE Xplore* – "[ID of Mosquitoes Using Wing Beats Sound on TinyML Embedded System](#)", and won 1st place, ITU International Hackathon. Developed a novel embedded AI system that classifies mosquito species by analyzing their wingbeat frequencies using TinyML, enabling real-time, low-power vector surveillance for disease control in resource-limited settings. This work was recognized internationally for its innovation in applying edge AI to global health challenges.
- Published a peer-reviewed paper in the *Journal of Agriculture and Food Research* on "[FilletCam AI: fillet color profiling tool](#)". Designed and trained a deep learning model to assess fish fillet color quality from images, achieving high accuracy and enabling rapid, objective evaluation in aquaculture and food safety workflows. This research demonstrates the practical impact of AI in automating quality control for the seafood industry.